

```

    }
    // Validate Packer template
    stage('Template Validation') {
        steps {
            withCredentials([string(
credentialsId: 'AWS_ACCESS_KEY_ID',
variable: 'AWS_ACCESS_KEY_ID'),
string(
credentialsId: 'AWS_SECRET_ACCESS_KEY',
variable: 'AWS_SECRET_ACCESS_KEY')]) {
                sh script: 'cd
packer/webserver/ && /opt/tools/packer
validate -var-file=vars.json app.json',
returnStatus: true
            }
        }
    }
    // Build Golden Image
    stage('Build Golden Image') {
        steps {
            withCredentials([string(
credentialsId: 'AWS_ACCESS_KEY_ID',
variable: 'AWS_ACCESS_KEY_ID'),
string(
credentialsId: 'AWS_SECRET_ACCESS_KEY',
variable: 'AWS_SECRET_ACCESS_KEY')]) {
                sh script: 'cd
packer/webserver/ && /opt/tools/packer
build -var-file=vars.json app.json',
returnStatus: true
            }
        }
    }
    // Validate Terraform Templates
    stage('Validate') {
        steps {
            withCredentials([string(
credentialsId: 'AWS_ACCESS_KEY_ID',
variable: 'AWS_ACCESS_KEY_ID'),
string(
credentialsId: 'AWS_SECRET_ACCESS_KEY',
variable: 'AWS_SECRET_ACCESS_KEY')]) {
                sh script: 'cd
terraform/webserver/ && /opt/tools/
terraform plan -auto-approve',
returnStatus: true
            }
        }
    }
    // Deploy golden ami to AWS
    cloud
    stage('Deploy') {

```

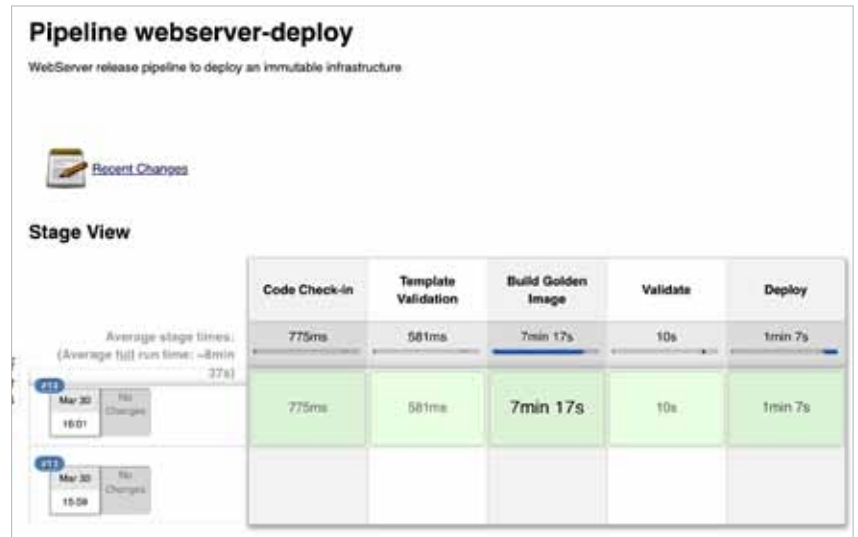


Figure 2: Jenkins pipeline



Figure 3: Final output

```

        steps {
            withCredentials([string(
credentialsId: 'AWS_ACCESS_KEY_ID',
variable: 'AWS_ACCESS_KEY_ID'),
string(
credentialsId: 'AWS_SECRET_ACCESS_KEY',
variable: 'AWS_SECRET_ACCESS_KEY')]) {
                sh script: 'cd
terraform/webserver/ && /opt/tools/
terraform apply -auto-approve,
returnStatus: true
            }
        }
    }
}

```

Finally, trigger the pipeline job, and wait for it to finish. If everything goes well, as shown in Figure 2, the application instance should be up and running in the AWS cloud (Figure 3).

The benefits of immutable infrastructure are not just restricted to faster deployments, safer rollbacks, enhanced security and scaling, but allow service owners like SysOps, SecOps, SREs, developers, etc, to move away from older, time-consuming ways of managing servers. **END** 🐧

References

- [1] <https://github.com/vijayg92/immutable-infrastructure>
- [2] <https://martinfowler.com/bliki/ImmutableServer.html>
- [3] <https://www.hashicorp.com/resources/what-is-mutable-vs-immutable-infrastructure>
- [4] <https://eplexity.com/blog/benefits-of-immutable-infrastructure/>

By: Vijay Gosai

The author is an open source enthusiast and is working as a senior site reliability engineer. His interests include distributed systems, cloud computing, infrastructure automation, software development, design patterns, DevOps and emerging technologies.